



Name: _____

Teacher: _____

Class: _____

FORT STREET HIGH SCHOOL

2006

PRELIMINARY SCHOOL CERTIFICATE COURSE

ASSESSMENT TASK 3: FINAL PRELIMINARY EXAMINATION

Mathematics Extension I

TIME ALLOWED: 2 HOURS

Outcomes Assessed	Questions	Marks
Demonstrates appropriate mathematical techniques in basic algebra, equations and geometry.	1, 2	
Manipulates algebraic expressions to solve problems from topic areas such as functions, trigonometry and quadratics.	3, 5	
Demonstrates skills in the processes of calculus and applies them appropriately.	4, 6	
Synthesises mathematical solutions to harder problems and communicates them in appropriate form.	7	

Question	1	2	3	4	5	6	7	Total	%
Marks	/12	/12	/12	/12	/12	/12	/12	/84	

Directions to candidates:

- Attempt all questions
- The marks allocated for each question are indicated on the right thus: [1]
- All necessary working should be shown in every question. Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used
- Each new question is to be started in a new booklet

Question One: 12 marks (Start a new booklet)

a) Express $\frac{1+\sqrt{8}}{1-3\sqrt{2}}$ with a rational denominator. [2]

b) If $H = (1-k)^n$, find the value of k (to 4 significant figures) given $H = 0.13$ and $n = 7$ [2]

c) Express $\frac{1-x^{-1}}{x^{-1}-x^{-2}}$ in its simplest form [2]

d) Find the values of θ , with $0^\circ \leq \theta \leq 360^\circ$, for which $2\cos\theta = -\sqrt{2}$ [2]

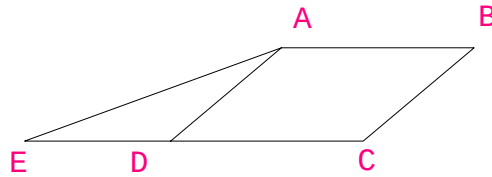
e) Find the value of $\frac{a^2b^3}{c^2}$ given $a = \left(\frac{3}{4}\right)^3$, $b = \left(\frac{2}{3}\right)^2$ and $c = \left(\frac{1}{2}\right)^4$ as a fraction. [2]

f) Sketch the graph of $y = |2x - 1|$ [2]

Question Two: 12 marks (Start a new booklet)

a) Solve for x : $\frac{2}{x} > x - 1$. [3]

b) ABCD is a parallelogram with CD produced to E so that ED = AD.



Prove that $\angle ABC = 2\angle DEA$. [2]

c) Given $\log 2 = a$ and $\log 3 = b$, find $\log 2.25$ in terms of a and b . [2]

d) For the lines $y = 3x - 1$ and $x - 2y + 5 = 0$:

i) Find the acute angle between these lines. [2]

ii) Using the k -method or otherwise, find the equation of the line through the intersection of these lines that passes through $P(1, -1)$ [3]

Question Three: 12 marks (Start a new booklet)

- a) The angle of depression from the top of a 120m cliff down to a boat in line with the bottom of the cliff is 35° . Calculate the distance of the boat from the base of the cliff (to the nearest m). [2]
- b) The quadratic equation $3x^2 - 5x + 2 = 0$ has roots α and β . Find the value of
- i) $(\alpha - 1)(\beta - 1)$ [1]
- ii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ [2]
- c) Find the coordinates of the point that divides the interval between $A(1, -6)$ and $B(-4, 3)$ externally in the ratio $3 : 2$. [2]
- d) Solve $x^4 - 3x^2 - 4 = 0$ for x . [3]
- e) Prove $\frac{\cos \theta (\sin \theta + \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)} = 1 + \tan \theta$ [2]

Question Four: 12 marks (Start a new booklet)

a) Differentiate with respect to x :

i) $x^3 - 5x^2 + 3$ [1]

ii) $(5 - 2x^2)^7$ [1]

iii) $\frac{2x}{1 + 4x}$ [2]

b) Find the gradient function for $f(x) = \frac{1}{x}$ from first principles. [3]

c) For the curve $y = 4 + 3x - x^2$, find the:

i) equation of the tangent at $x = 1$ (in General Form) [3]

ii) find the exact distance of point $P(2, -1)$ from this tangent [2]

Question Five: 12 marks (Start a new booklet)

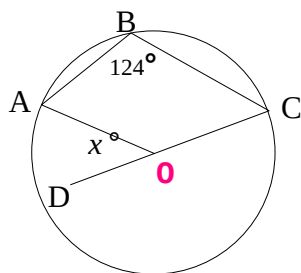
- a) Find the values of a , b and c such that $2x^2 - 5x + 7$ can be expressed in the form $a(x+b)^2 + c$

[4]

- b) Find the value of x , giving reasons.

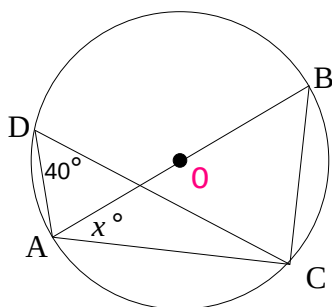
i)

[2]



ii)

[2]



- c) For what values of k does the quadratic $x^2 + (k+1)x - (k-1) = 0$ have distinct real roots?

[4]

Question Six: 12 marks (Start a new booklet)

a) Points $P(2,4)$ and $Q(x_1, y_1)$ are on the parabola $y = x^2$:

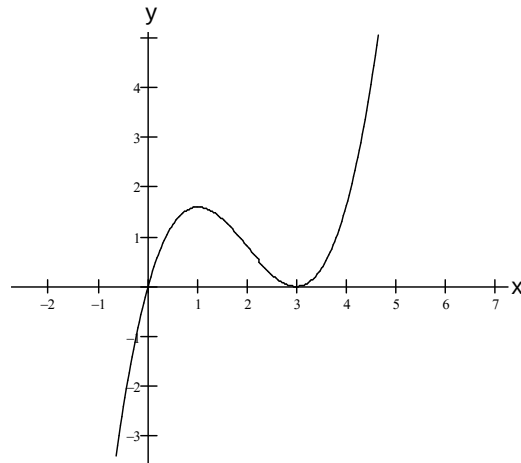
i) Show the equation of the normal at point P is given by $x + 4y - 18 = 0$.

[1]

ii) The tangent at Q is parallel to the normal at P . Find the coordinates of Q and the equation (in general form) of the tangent at Q .

[3]

b) For the function $y = f(x)$, the graph of its derivative $y = f'(x)$ is shown below.



i) For what values of x is $f(x)$ rising?

[1]

ii) What are the x -values of the stationary points of $f(x)$?

[1]

iii) Given $x = -1$ is a root of $f(x)$, and the minimum value of $f(x)$ is -3 , draw a neat sketch of a possible $y = f(x)$ for $-2 < x < 5$.

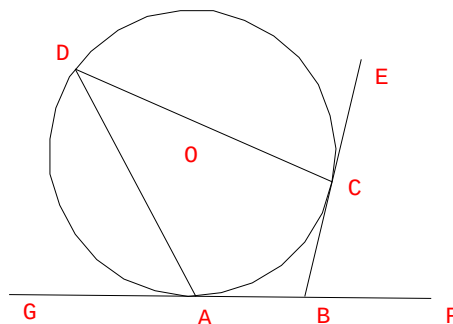
[2]

c) For the curve given by $y = x^3 + 3x^2 - 9x - 7$, find any stationary points or points of inflection and determine their nature.

[4]

Question Seven: 12 marks (Start a new booklet)

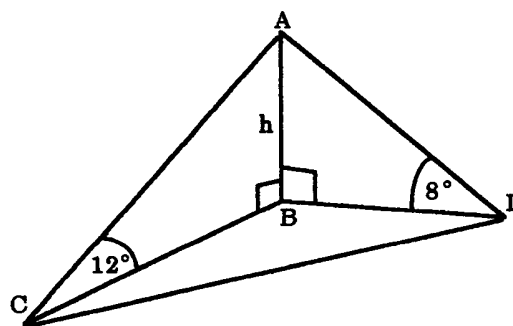
- a) A and C are two separate points on the circle, centre O, as shown in the diagram below. Tangents at A and C meet at B. D is a point on the circumference of the circle.



Prove that quadrilateral ABCD can never be cyclic.

[4]

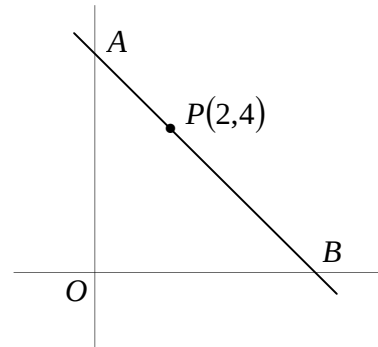
- b) AB is a tower of height h metres. From points C and D in the same plane as the base of the tower, the angles of elevation to the top of the tower are 12° and 8° respectively, as shown in the diagram opposite. From the base of the tower, the bearing of points C and D are $229^\circ T$ and $187^\circ T$ respectively. Find the height of the tower (to the nearest metre) if C is 400m from D.



[3]

(Question 7 continues over)

- c) A straight line passing through the point $P(2,4)$ cuts the positive axes at A and B.



- i) Show the equation of the line is $y = mx - 2m + 4$, where m is the gradient of the line. [1]
- ii) Show that points A and B, where the line meets the axes, have coordinates $(0, 4 - 2m)$ and $\left(\frac{2m - 4}{m}, 0\right)$ respectively. [1]
- iii) Find the value of m for which the area of triangle AOB is least. [3]

Solutions: Preliminary Final Exam
Extension 1 2006.

Question One:

$$\begin{aligned} \text{a)} \quad &= \frac{1-\sqrt{8}}{1-3\sqrt{2}} \times \frac{1+3\sqrt{2}}{1+3\sqrt{2}} \\ &= \frac{(1-2\sqrt{2})(1+3\sqrt{2})}{(1-18)} \\ &= \frac{1-2\sqrt{2}+3\sqrt{2}-12}{17} \\ &= \frac{3\sqrt{2}-11}{17} \end{aligned}$$

$$\begin{aligned} \text{b)} \quad &H = (1-k)^n \\ \text{i.e. } &\sqrt[3]{0.13} = 1-k \\ \text{or } &k = 1 - \sqrt[3]{0.13} \\ \therefore &k = 0.252828189 \\ \text{i.e. } &k = 0.2528 \text{ (to 4 sig. figs)} \end{aligned}$$

$$\begin{aligned} \text{c)} \quad &\frac{1-x^{-1}}{x^{-1}-x^{-2}} \\ &= \frac{1-\frac{1}{x}}{\frac{1}{x}-\frac{1}{x^2}} \\ &= \frac{\frac{x-1}{x}}{\frac{x-1}{x^2}} \\ &= \frac{x-1}{x} \times \frac{x^2}{x-1} \\ &= x \end{aligned}$$

$$\begin{aligned} \text{d)} \quad &2\cos\theta = -\sqrt{2} \\ \cos\theta &= \frac{-\sqrt{2}}{2} \\ \theta &= \cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) \end{aligned}$$

i.e. base ϕ is 45°
 Cos is negative in Q's 2 & 3

$$\therefore \theta = 180 - \phi, 180 + \phi$$

$$\text{i.e. } \theta = 135^\circ, 225^\circ \quad [1]$$

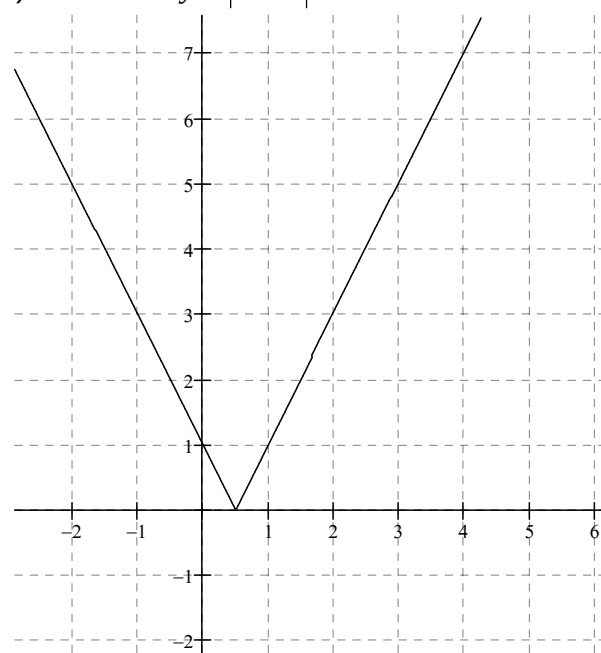
$$\begin{aligned} \text{e)} \quad &a = \left(\frac{3}{4}\right)^3, b = \left(\frac{2}{3}\right)^2, c = \left(\frac{1}{2}\right)^4 \\ \therefore &\frac{a^2 b^3}{c^2} \end{aligned}$$

$$\begin{aligned} &= \frac{\left(\left(\frac{3}{4}\right)^3\right)^2 \left(\left(\frac{2}{3}\right)^2\right)^3}{\left(\left(\frac{1}{2}\right)^4\right)^2} \end{aligned}$$

$$= \frac{3^6}{4^6} \times \frac{2^6}{3^6} \div \frac{1}{2^8} \quad [1]$$

$$\begin{aligned} &= \frac{3^6 \cdot 2^6 \cdot 2^8}{2^{12} \cdot 3^6} \\ &= 2^2 \\ &= 4 \end{aligned} \quad [1]$$

f) Sketch $y = |2x - 1|$



"Vertex" correct [1]
 Shape with correct y intercept [1/2]

Question Two:

a) $x^2 : 2x > x^3 - x^2$

i.e. $x^3 - x^2 - 2x < 0$

$\therefore x(x^2 - x - 2) < 0$

Boundary points when

$x(x^2 - x - 2) = 0$

$x(x-2)(x+1) = 0$

i.e. $x = -1, 0, 2$ [1]

so for $\frac{2}{x} > x-1$

testing points between boundaries: [1]

$x = -2; x = \frac{-1}{2}; x = 1; x = 3$

LHS: LHS: LHS: LHS:

$= \frac{2}{-2} = \frac{2}{\frac{-1}{2}} = \frac{2}{1} = \frac{2}{3}$

$= -1 = -4 = 2$

RHS: RHS: RHS: RHS:

$= -2 - 1 = \frac{-1}{2} - 1 = 1 - 1 = 3 - 1$

$= -3 = -1\frac{1}{2} = 0 = 2$

<LHS >LHS <LHS >LHS

False True False True

$\therefore -1 < x < 0, x > 2$ is soln. [1]

(any equivalent method ok)

b) Let $\angle DEA = x$

$\therefore \angle EAD = x$ (base $\angle$'s Isos Δ)

$\therefore \angle CDA = x + x$ (ext. $\angle$ of ΔEDA) [1]

$= 2x$

$\therefore \angle ABC = 2x$ (op. $\angle$'s ||gram =) [1]

$\therefore \angle ABC = 2\angle DEA$

c) $\log 2.25$

$= \log \frac{9}{4}$

$= \log \frac{3^2}{2^2}$

$= \log 3^2 - \log 2^2$ [1]

$= 2\log 3 - 2\log 2$

$= 2a + 2b$ [1]

d) $y = 3x - 1$ and $x - 2y + 5 = 0$

i) $m_1 = 3$ and $m_2 = \frac{1}{2}$, so

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{3 - \frac{1}{2}}{1 + 3 \times \frac{1}{2}} \right|$ [1]

$= \left| \frac{2\frac{1}{2}}{2\frac{1}{2}} \right|$

$= 1$

$\therefore \theta = \tan^{-1}(1)$

$= 45^\circ$ [1]

ii) Required equation is given by

$3x - y - 1 + k(x - 2y + 5) = 0$ [1]

Substituting for $P(1, -1)$

$3 \times 1 - (-1) - 1 + k(1 - 2 \times (-1) + 5) = 0$

i.e. $3 + k(8) = 0$

So $k = \frac{-3}{8}$ [1]

$\therefore$ equation is

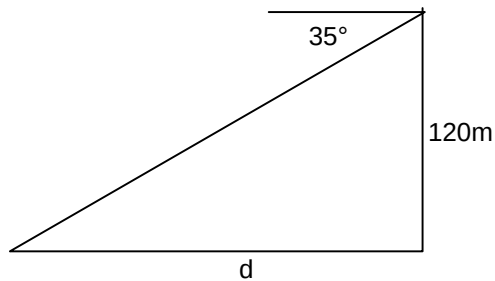
$3x - y - 1 + \left(\frac{-3}{8} \right) (x - 2y + 5) = 0$

$24x - 8y - 8 + (-3x + 6y - 15) = 0$

$\therefore 21x - 2y - 23 = 0$ [1]

Question Three:

a)



$$\tan 35 = \frac{120}{d} \quad [1]$$

$$d = \frac{120}{\tan 35}$$

$$d = 171.3777608$$

$$d = 171m \quad [1]$$

b) $\alpha + \beta = \frac{5}{3}; \alpha\beta = \frac{2}{3}$

i) $(\alpha - 1)(\beta - 1)$

$$= \alpha\beta - (\alpha + \beta) + 1$$

$$= \frac{5}{3} - \frac{2}{3} + 1$$

$$= 2 \quad [1]$$

ii) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$

$$= \frac{\alpha^2 + \beta^2}{\alpha^2 \beta^2}$$

$$= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{(\alpha\beta)^2} \quad [1]$$

$$= \frac{\left(\frac{5}{3}\right)^2 - 2 \times \frac{2}{3}}{\left(\frac{2}{3}\right)^2}$$

$$= \frac{(25 - 12)/9}{4/9}$$

$$= \frac{13}{4}$$

$$= 3\frac{1}{4} \quad [1]$$

c) $A(1, -6)$ and $B(-4, 3)$

Ratio $-3 : 2$ (-ve for external)

$$\left(\frac{1 \times 2 + (-4) \times (-3)}{-3 + 2}, \frac{(-6) \times 2 + 3 \times (-3)}{-3 + 2} \right) \quad [1]$$

$$= \left(\frac{14}{-1}, \frac{-21}{-1} \right)$$

$$= (-14, 21) \quad [1]$$

d) Solve $x^4 - 3x^2 - 4 = 0$

Let $u = x^2$

$$\therefore u^2 - 3u - 4 = 0 \quad [1]$$

$$(u + 1)(u - 4) = 0$$

i.e. $u = -1, 4$

$$\therefore x^2 = -1 \text{ (no real solution)} \quad [1]$$

or $x^2 = 4$

$$\therefore x = \pm 2 \quad [1]$$

e) Prove

$$\frac{\cos \theta (\sin \theta + \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)} = 1 + \tan \theta$$

$$\text{LHS} = \frac{\cos \theta (\sin \theta + \cos \theta)}{(1 + \sin \theta)(1 - \sin \theta)}$$

$$= \frac{\cos \theta (\sin \theta + \cos \theta)}{(1 - \sin^2 \theta)}$$

$$= \frac{\cos \theta (\sin \theta + \cos \theta)}{\cos^2 \theta} \quad [1]$$

$$= \frac{(\sin \theta + \cos \theta)}{\cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\cos \theta}$$

$$= \tan \theta + 1$$

$$= \text{RHS}$$

Question Four:

a) Differentiations:

i) $3x^2 - 5$ [1]

ii) $7(5 - 2x^2)^6(-4x)$ [1]
 $= -28x(5 - 2x^2)^6$

iii) $\frac{2x}{1+4x}$; $u = 2x, v = 1 + 4x$ [1]
 $du = 2; dv = 4$
 $\therefore y' = \frac{(1+4x) \times 2 - 2x \times 4}{(1+4x)^2}$ [1]
 $= \frac{2+8x-8x}{(1+4x)^2}$
 $= \frac{2}{(1+4x)^2}$ [1]

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$= \frac{|2 - (-1) + 5|}{\sqrt{1^2 + (-1)^2}}$$

$$= \frac{|8|}{\sqrt{2}}$$

$$= \frac{8\sqrt{2}}{2}$$

$$= 4\sqrt{2}$$
 [1]

b) $f(x) = \frac{1}{x}$ and $f(x+h) = \frac{1}{x+h}$ [1]
 $\therefore \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
 $= \lim_{h \rightarrow 0} \frac{\frac{1}{x+h} - \frac{1}{x}}{h}$ [1]
 $= \lim_{h \rightarrow 0} \frac{\frac{x - (x+h)}{x(x+h)}}{h}$
 $= \lim_{h \rightarrow 0} \frac{-h}{hx(x+h)}$ [1]
 $= \lim_{h \rightarrow 0} \frac{-1}{x^2 + hx}$
 $= \frac{-1}{x^2}$ [1]

c) $y = 4 + 3x - x^2$

i) $y' = 3 - 2x$ [1]

so for tangent at $x = 1$: $y' = 1$
 and $y = 6$ [1]

$\therefore y - 6 = 1(x - 1)$
 or $0 = x - y + 5$ [1]

ii) For distance of $P(2, -1)$ from tangent:

Question Five:

a) $a(x+b)^2 + c$
 $= ax^2 + 2abx + b^2 + c$
i.e.
 $2x^2 - 5x + 7 \equiv ax^2 + 2abx + b^2 + c$ [1]
 $\therefore a = 2$ [1]
also $2ab = -5$
 $\therefore b = -\frac{5}{4}$ [1]
and $b^2 + c = 7$
 $\therefore c = 7 - \left(-\frac{5}{4}\right)^2$
so $c = \frac{87}{16}$
or $c = 5\frac{7}{16}$ [1]

b) i) Reflex $\angle AOC = 2 \times 124^\circ$
(angle at centre double angle at circumference) [1]
 $\therefore \angle AOC = 248^\circ$
but $\therefore \angle AOC = x^\circ + \angle DOC$
where $\angle DOC$ is a st.line.
 $\therefore 248 = x + 180$
i.e. $x = 68$ [1]
ii) $\angle ADC = \angle ABC$ (in same seg.
Standing on same arc)
 $\therefore \angle ABC = 40^\circ$
 $\angle ACB = 90^\circ$ (angle in semicircle) [1]
 $\therefore 40 + 90 + x = 180$ (Δ sum)
so $x = 50$ [1]

c) For $x^2 + (k+1)x - (k-1) = 0$
Distinct real roots when $\Delta > 0$.
i.e. $(k+1)^2 - 4 \times 1 \times -(k-1) > 0$
 $k^2 - 2k + 1 + 4k - 8 > 0$
 $\therefore k^2 + 2k - 7 > 0$ [1]
Boundaries when $\Delta = 0$
i.e. when $k^2 + 2k - 7 = 0$
 $\therefore k = \frac{-(2) \pm \sqrt{2^2 - 4 \times 1 \times -7}}{2 \times 1}$

$$k = \frac{-2 \pm \sqrt{32}}{2}$$

$$k = \frac{-2 \pm \sqrt{32}}{2}$$

$$k = \frac{-2 \pm 4\sqrt{2}}{2}$$

$$k = -1 \pm 2\sqrt{2}$$
 [1]

Testing $k=0$ in $k^2 + 2k - 7 > 0$
LHS = -7
< 0
< RHS
i.e. false so 0 is not a solution. [1]
 $\therefore k < -1 - 2\sqrt{2}$
or $k > -1 + 2\sqrt{2}$ [1]

Question Six:

a) $P(2,4)$ and $Q(x_1, y_1)$ on $y = x^2$
 $\frac{dy}{dx} = 2x$
i) At $P(2,4)$:
 $\frac{dy}{dx} = 4$
So gradient of normal is $-\frac{1}{4}$:
 $\therefore (y-4) = -\frac{1}{4}(x-2)$ [1]
 $4y - 16 = -x + 2$
i.e. $x + 4y - 18 = -x + 2$
ii) At $Q(x_1, y_1)$, $\frac{dy}{dx} = 2x_1$ but also
 $\frac{dy}{dx} = -\frac{1}{4}$
 $\therefore 2x_1 = -\frac{1}{4}$ [1]
i.e. $x_1 = -\frac{1}{8}$

$$\therefore y_1 = \frac{1}{64} \text{ so } Q \text{ is } \left(\frac{-1}{8}, \frac{1}{64} \right) \quad [1]$$

Equation of the tangent is:

$$\left(y - \frac{1}{64} \right) = \frac{-1}{4} \left(x - \frac{-1}{8} \right)$$

$$4y - \frac{1}{16} = -x - \frac{1}{8}$$

$$\therefore 64y - 1 = -16x - 2$$

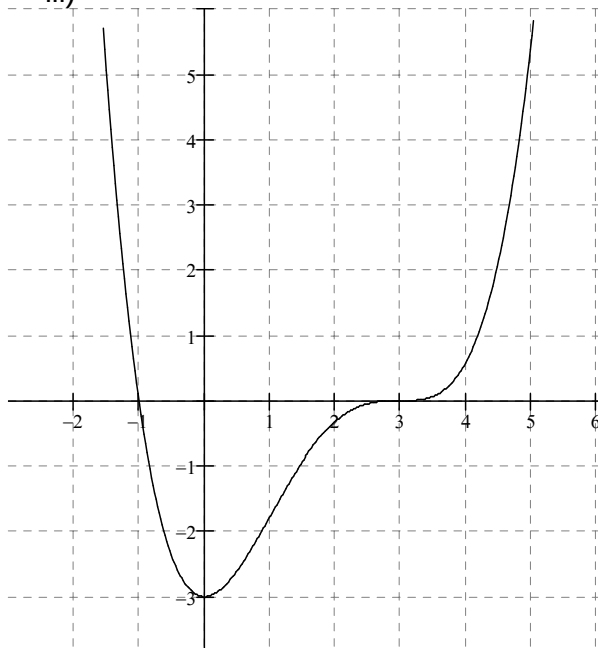
$$\text{or } 16x + 64y + 1 = 0 \quad [1]$$

b)

i) $f(x)$ is rising when $f'(x) > 0$
i.e. for $x > 0$ [1]

ii) Stat points when $f'(x) = 0$
i.e. when $x = 0$ or $x = 3$ [1]

iii)



Correct roots [1]

Correct shape [1]

c) $y = x^3 + 3x^2 - 9x - 7$

$$\therefore y' = 3x^2 + 6x - 9$$

$$\text{and } y'' = 6x + 6$$

Stat Points when $y' = 0$:

$$\therefore 0 = 3x^2 + 6x - 9$$

$$\text{or } 0 = x^2 + 2x - 3$$

$$\text{i.e. } 0 = (x + 3)(x - 1)$$

$$\therefore x = -3 \text{ or } x = 1$$

y-values are:

$$x = -3$$

$$y = (-3)^3 + 3(-3)^2 - 9(-3) - 7$$

$$y = 20, \text{ so point is } (-3, 20)$$

$$x = 1$$

$$y = (1)^3 + 3(1)^2 - 9(1) - 7$$

$$y = -12, \text{ so point is } (1, -12)$$

Nature:

$$x = -3: y'' = 6(-3) + 6$$

i.e. $y'' = -12$ so ccd $\Rightarrow$ max t.p.

$$x = 1: y'' = 6(1) + 6$$

i.e. $y'' = 12$ so ccu $\Rightarrow$ min t.p.

Possible inflection point(s) when

$$y'' = 0:$$

$$\text{i.e. } 0 = 6x + 6$$

$$\text{so } x = -1$$

y-value is:

$$y = (-1)^3 + 3(-1)^2 - 9(-1) - 7$$

$$y = 4, \text{ so point is } (-1, 4)$$

Testing:

$$\text{For } x = -1 - \varepsilon$$

$$y'' = 6(-1 - \varepsilon) + 6$$

$$y'' = -6\varepsilon$$

$$y'' < 0$$

$$\text{For } x = -1 + \varepsilon$$

$$y'' = 6(-1 + \varepsilon) + 6$$

$$y'' = 6\varepsilon$$

$$y'' > 0$$

$\therefore$ as concavity changes, $(-1, 4)$ is a point of inflection.

Marking:

Finds correct Stat points [1]

Nature tests correct [1]

Finds Inflection point [1]

Tests nature correctly [1]

Question Seven:

a) Join AC

$$\text{Let } \angle CAB = x^\circ$$

Now $AB = BC$ (tangents from external point equal)

$\therefore \triangle ABC$ is isosceles

$$\therefore \angle ACB = x^\circ \text{ (base } \angle \text{'s Isos } \Delta)$$

$$\therefore \angle ABC = (180 - 2x)^\circ \text{ (}\Delta \text{ sum)}$$

Also, $\angle ADC = x^\circ$ (angle in alt. seg.)

For $ABCD$ to be cyclic, opposite angles must be supplementary;

$$\text{i.e. } \angle ADC + \angle ABC = 180^\circ$$

$$\text{i.e. } x + (180 - 2x) = 180$$

$$\text{which gives } x = 0$$

but A and C are separate points,
so $x > 0$

$\therefore ABCD$ cannot be cyclic.

b) $\angle CBD = 229^\circ - 187^\circ$

$$\therefore \angle CBD = 42^\circ$$

$$\text{Also, } \frac{h}{BD} = \tan 8^\circ$$

$$\therefore BD = h \cot 8^\circ$$

$$\text{and } \frac{h}{BC} = \tan 12^\circ$$

$$\therefore BC = h \cot 12^\circ$$

(correct expressions for BC , BD)

In ΔCBD :

$$CD^2 = BD^2 + BC^2 - 2 \cdot BC \cdot BD \cdot \cos \angle CBD$$

$$400^2 = (h \cot 8^\circ)^2 + (h \cot 12^\circ)^2 - 2 \times h \cot 8^\circ \times h \cot 12^\circ \times \cos 42^\circ$$

(correct substitutions)

$$h^2 = \frac{400^2}{\cot^2 8^\circ + \cot^2 12^\circ - 2 \cot 8^\circ \cot 12^\circ \cos 42^\circ}$$

$$\therefore h^2 = \frac{160000}{23.00821324}$$

$$h^2 = 6954.03847$$

$$\therefore h = 83.39087762$$

$$\therefore h = 83 \text{ (nearest m)}$$

c) $P(2,4)$

i) $(y - y_1) = m(x - x_1)$ with $P(2,4)$
gives:

$$(y - 4) = m(x - 2)$$

$$y - 4 = mx - 2m$$

$$\text{or } y = mx - 2m + 4$$

ii) For A , $x = 0$, subst in (i) gives

$$y = -2m + 4$$

$$\therefore A(0, 4 - 2m)$$

For B , $y = 0$, subst in (i) gives

$$0 = mx - 2m + 4$$

$$\text{or } mx = 2m - 4$$

$$\text{so } x = \frac{2m - 4}{m}$$

$$\therefore B\left(\frac{2m - 4}{m}, 0\right)$$

(Correctly shown)

iii) Area of ΔAOB is given by

$$A = \frac{1}{2}bh \text{ where}$$

$$b = \frac{2m - 4}{m} \text{ and } h = 4 - 2m$$

$$\therefore A = \frac{1}{2} \times \frac{(2m - 4)}{m} \times (4 - 2m)$$

$$\text{i.e. } A = \frac{-16 + 16m - 4m^2}{2m}$$

$$\text{or } A = 8 - \frac{8}{m} - 2m$$

$$\therefore \frac{dA}{dm} = \frac{8}{m^2} - 2$$

$$\text{and } \frac{d^2A}{dm^2} = \frac{-16}{m^3}$$

$$\text{Min/Max when } \frac{dA}{dm} = 0$$

$$\text{i.e. } 0 = \frac{8}{m^2} - 2$$

$$\text{or } 0 = 8 - 2m^2$$

$$\therefore 2m^2 = 8$$

$$m^2 = 4$$

$$\text{so } m = \pm 2$$

$$\text{at } m = 2$$

$$\frac{d^2A}{dm^2} = \frac{-16}{2^3}$$

$$< 0 \text{ ccd} \Rightarrow \text{max t.p.}$$

$$\text{at } m = -2$$

$$\frac{d^2A}{dm^2} = \frac{-16}{(-2)^3}$$

$$> 0 \text{ ccu} \Rightarrow \text{min t.p.}$$

$\therefore m = -2$ is the gradient that gives the least area for ΔAOB